Lecture 18 Scribe

L18.pdf
PDF

- ALTERNATE HIGH-PRECISION LECTURE SCRIBE PROMPT

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▪ PRIMARY OBJECTIVE

Convert the given lecture into a document that is:

Complete in conceptual coverage

Mathematically exact

Structurally organized

Optimized for rapid exam revision

This is NOT a summary task. You must reconstruct the lecture in full detail, preserving all technical depth and logical rigor.

▪ CONTENT CONSTRAINTS (MANDATORY)

Use strictly and exclusively the provided lecture content

Do NOT introduce:

External definitions

Additional examples

Alternative explanations

Do NOT omit:

Any formula

Any derivation

Any intermediate reasoning step

Maintain exact notation, symbols, and structure as given

Rephrase only for clarity — never simplify away rigor

Preserve all references to equations, tables, and figures (including slide-based structure)

▪ WRITING STYLE REQUIREMENTS

Formal academic tone (no conversational language)

No emojis, stylistic decorations, or informal markers

No storytelling, intuition, or analogies unless explicitly present in slides

Precise, concise, and structured exposition

Maintain consistency in formatting and hierarchy

▪ MATHEMATICAL FORMATTING RULES

All formulas must be presented in boxed plain-text format, exactly as:

```
|-------------------------------|
| Expression / Formula Here     |
|-------------------------------|
```

Guidelines:

Keep formulas visually isolated from text

Preserve original notation exactly

Group related formulas logically

Categorize clearly:

Core definitions

Derived expressions

Properties

▪ DOCUMENT STRUCTURE
1. Lecture Overview

Enumerate all topics covered

Reflect the exact progression of the lecture

2. Topic-wise Detailed Reconstruction

For each topic:

(A) Definitions and Concepts

State all definitions exactly

Maintain original terminology

(B) Formula Set

Include ALL formulas (boxed)

Preserve ordering and grouping

(C) Derivations

Present step-by-step derivations

No skipped steps

Each transformation must follow logically from the previous

(D) Properties and Results

List all properties explicitly

Include conditions and assumptions

(E) Problem-Solving Methodology

Extract the exact solving approach demonstrated in lecture

Present as structured steps

3. Worked Examples

For every example:

State the problem exactly

Identify the concept being applied

Show complete step-by-step solution

Include every intermediate calculation

Clearly present final result

Also include:

Mistakes implied by the solution steps (if any)

Key logical takeaways

4. Final Revision Section

(A) Formula Compendium

All formulas only (boxed)

No explanations

(B) Core Concept Statements

One-line precise statements

(C) Standard Solution Patterns

Recognizable problem-solving templates

▪ REASONING REQUIREMENT

Expand every derivation and solution exactly as shown in the lecture

Do not compress multi-step logic into single statements

Do not add reasoning beyond what exists in the slides

▪ OUTPUT QUALITY CHECK

The final document must:

Fully cover the lecture (no missing content)

Be mathematically correct and logically consistent

Be usable as a standalone exam revision resource

Reflect the structure and rigor of actual lecture delivery

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Do NOT simplify.
Do NOT generalize.
Do NOT add external knowledge.

Produce a complete, structured, exam-ready lecture scribe based solely on the provided material.

▪ OUTPUT FORMAT

Return the final output, properly structured with:

Sections and subsections

Equation environments

Clean formatting suitable for academic submission

Below is the **FORMAL, EXAM-ORIENTED Lecture Scribe** reconstructed strictly from the provided Lecture 18 slides [L18].

Lecture 18: Marginal PMF, Correlation, Covariance, Joint Gaussian RVs

---

1. Lecture Overview

The lecture covers:

1. Marginal PMF
    - Definition
    - Computation from joint PMF table
    - Solved example
2. Correlation
    - Product moment definition
    - Interpretation
3. Covariance
    - Definition and interpretation
    - Algebraic form
    - Properties
    - Solved examples
    - Pearson's coefficient
4. Joint Gaussian Random Variables
    - Definition
    - Algebraic form
    - Interpretation

---

2. Topic-wise Detailed Reconstruction

2.1 Marginal PMF

(A) Definitions and Concepts

For discrete random variables X and Y:

|----------|

$$p_{XY}(x, y) = P(X = x, Y = y)$$

The marginal PMFs are obtained by summing over the other variable:

|----------|

$$p_X(x) = \sum_y p_{XY}(x, y)$$

|----------|

$$p_Y(y) = \sum_x p_{XY}(x, y)$$

Key Idea:
The marginal PMF of one variable is obtained by summing the joint PMF over all possible values of the other variable. *(Slide 3)*

(B) Formula Set

|----------|

$$p_X(x_i) = \sum_j p_{ij}$$

|----------|

$$p_Y(y_j) = \sum_i p_{ij}$$

|----------|

$$\sum_i p_X(x_i) = 1$$

|----------|

$$\sum_j p_Y(y_j) = 1$$

(C) Derivation

From joint PMF table (Slide 4):

- Row sums $\rightarrow p_X(x)$
- Column sums $\rightarrow p_Y(y)$

(D) Properties

- Marginals are valid PMFs
- Sum to unity

(E) Problem-Solving Methodology

1. Identify joint PMF table
2. Sum across rows $\rightarrow$ obtain $p_X(x)$
3. Sum across columns $\rightarrow$ obtain $p_Y(y)$
4. Verify normalization

(F) Worked Example

Given (Slide 5)

Joint PMF:

x \ y	0	1
0	1/8	1/4
1	3/8	1/4

Solution

Compute marginals:

|----------|

$$p_X(0) = \frac{1}{8} + \frac{1}{4} = \frac{3}{8}$$

|----------|

$$p_X(1) = \frac{3}{8} + \frac{1}{4} = \frac{5}{8}$$

|----------|

$$p_Y(0) = \frac{1}{8} + \frac{3}{8} = \frac{1}{2}$$

|----------|

$$p_Y(1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

Final Result

|----------|

$$p_X(x) = \{3/8, 5/8\}$$

|----------|

$$p_Y(y) = \{1/2, 1/2\}$$

Key Takeaway

- Marginals are obtained via summation over the other variable.

2.2 Correlation

(A) Definition

|----------|

$$R_{XY} = E[XY]$$

(B) Interpretation

- Measures average product of X and Y
- Computed about origin (not means)
- Combines:
  - Mean
  - Spread
  - Dependence

(C) Properties

- Depends on scale of variables
- $R_{XY} = 0 \Rightarrow$ orthogonal about origin
- Not a clean measure of dependence

(D) Transition

To remove mean effects → **Covariance**

2.3 Covariance

(A) Definition

|----------|

$$\mathrm{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

Where:

|----------|

$$\mu_X = E[X], \quad \mu_Y = E[Y]$$

(B) Algebraic Form (Derivation)

Start:

|----------|

$$E[(X - \mu_X)(Y - \mu_Y)]$$

Expand:

|----------|

$$= E[XY - X\mu_Y - Y\mu_X + \mu_X\mu_Y]$$

Apply expectation:

|----------|

$$= E[XY] - \mu_Y E[X] - \mu_X E[Y] + \mu_X \mu_Y$$

Substitute means:

|----------|

$$= E[XY] - \mu_X \mu_Y$$

Final:

|----------|

$$\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

(C) Interpretation

- Positive: same direction variation
- Negative: opposite direction
- Zero: no linear relation

Magnitude depends on units *(Slide 9)*

(D) Properties

1. **Symmetry**

|----------|

$$\mathrm{Cov}(X, Y) = \mathrm{Cov}(Y, X)$$

2. **Variance as Special Case**

|----------|

$$\mathrm{Cov}(X, X) = \mathrm{Var}(X)$$

3. **Linearity**

|----------|

$$\mathrm{Cov}(aX + b, cY + d) = ac\,\mathrm{Cov}(X, Y)$$

4. **Independence**

|----------|

$$X \perp Y \Rightarrow \mathrm{Cov}(X, Y) = 0$$

Note: Zero covariance ≠ independence *(Slide 11)*

(E) Worked Example 1

Given:

|----------|

$$Y = -2X + 3$$

Solution

|----------|

$$\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

Substitute:

|----------|

$$XY = -2X^2 + 3X$$

|----------|

$$E[Y] = -2E[X] + 3$$

Final:

|----------|

$$\mathrm{Cov}(X, Y) = -2\mathrm{Var}(X)$$

Key Takeaway

- Linear transformations directly affect covariance.

(F) Worked Example 2

Given (Slide 14)

|----------|

$$f_{XY}(x,y) = \frac{xy}{96}, \quad 0 < x < 4,\ 1 < y < 5$$

Results

1.

|----------|

$$E[X] = \frac{8}{3}$$

2.

|----------|

$$E[Y] = \frac{31}{9}$$

3.

|----------|

$$E[XY] = \frac{248}{27}$$

4.

|----------|

$$E[2X + 3Y] = \frac{47}{3}$$

5.

|----------|

$$\mathrm{Var}(X) = \frac{8}{9}$$

6.

|----------|

$$\mathrm{Var}(Y) = \frac{92}{81}$$

7.

|----------|

$$\mathrm{Cov}(X, Y) = 0$$

8.

|----------|

$$\rho_{XY} = 0$$

Key Takeaway

- Independence ⇒ covariance = 0
- Leads to zero correlation

2.4 Pearson's Correlation Coefficient

(A) Definition

|----------|

$$\rho_{XY} = \frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}}$$

(B) Properties

- Measures strength and direction
- Normalized covariance
- Unit-free

(C) Range

|----------|

$$-1 \leq \rho_{XY} \leq 1$$

(D) Interpretation

- $\rho > 0$: positive linear trend
- $\rho < 0$: negative linear trend
- $\rho = 0$: no linear correlation

2.5 Joint Gaussian Random Variables

(A) Definition

A pair (X, Y) is jointly Gaussian if:

- Joint PDF is bivariate Gaussian
- Marginals are Gaussian *(Slide 22)*

(B) Algebraic Form

|----------|

$$f_{XY}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(\frac{-1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right)\right]\right)$$

(C) Parameters

- μ_X, μ_Y: means
- σ_X, σ_Y: standard deviations
- ρ: correlation coefficient

(D) Interpretation

- Models joint continuous behavior
- Widely used in:
  - Signal processing
  - Communication systems
  - Noisy measurements

(E) Example (Slide 23)

- X: CO_2 concentration
- Y: temperature anomaly

Joint Gaussian models their relationship due to multiple random influences.

---

3. Final Revision Section

(A) Formula Compendium

|----------|

$$p_X(x) = \sum_y p_{XY}(x, y)$$

|----------|

$$p_Y(y) = \sum_x p_{XY}(x, y)$$

|----------|

$$R_{XY} = E[XY]$$

|----------|

$$\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

|----------|

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2$$

|----------|

$$\rho_{XY} = \frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}}$$

|----------|

Joint Gaussian PDF (as above)

(B) Core Concept Statements

- Marginal PMF obtained by summation over joint PMF
- Correlation $E[XY]$ is origin-based
- Covariance centers variables
- Zero covariance ≠ independence
- Pearson coefficient normalizes covariance
- Joint Gaussian fully defined by mean, variance, correlation

---

(C) Standard Solution Patterns

1. **Marginal PMF**
   - Sum rows/columns
2. **Covariance**
   - Use $E[XY] - E[X]E[Y]$
3. **Linear Transformation**
   - Apply covariance linearity
4. **Expectation Problems**
   - Integrate using marginal PDFs
5. **Joint PDF Problems**
   - Compute marginals → expectations → variance → covariance → correlation

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End of Lecture Scribe

Sources

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Lecture 18: Marginal PMF, Correlation, Covariance, Joint Gaussian Random Variables

1. Lecture Overview

The lecture proceeds in the following order:

1. Marginal PMF
    - Definition
    - Marginal PMF from a Joint PMF Table
    - Solved Example
2. Correlation
    - Definition
    - Interpretation
3. Covariance
    - Definition and interpretation
    - Algebraic form and properties
    - Solved examples
    - Pearson's Coefficient
4. Joint Gaussian Random Variables
    - Definition and intuition
    - Algebraic form and interpretation

---

2. Topic-wise Detailed Reconstruction

2.1 Marginal PMF

(A) Definitions and Concepts

For discrete random variables X and Y, the joint PMF is:

|----------|

$$p_{XY}(x, y) = P(X = x, Y = y)$$

The marginal PMFs are obtained by summing over the other variable:

|----------|

$$p_X(x) = \sum_y p_{XY}(x, y)$$

|----------|

$$p_Y(y) = \sum_x p_{XY}(x, y)$$

Key Idea (Slide 3):
The marginal PMF of one variable is obtained by summing the joint PMF over all possible values of the other variable.

(B) Formula Set

From the joint PMF table representation:

|----------|

$$p_X(x_i) = \sum_j p_{ij}$$

|----------|

$$p_Y(y_j) = \sum_i p_{ij}$$

Normalization:

|----------|

$$\sum_i p_X(x_i) = 1$$

|----------|

$$\sum_j p_Y(y_j) = 1$$

(C) Derivations

From the joint PMF table (Slide 4):

- Row-wise summation:

|----------|

$$p_X(x_i) = \sum_j p_{ij}$$

- Column-wise summation:

|----------|

$$p_Y(y_j) = \sum_i p_{ij}$$

(D) Properties and Results

- Marginal PMFs satisfy normalization
- Derived directly from joint PMF
- Row sums correspond to $p_X(x)$
- Column sums correspond to $p_Y(y)$

(E) Problem-Solving Methodology

1. Identify the joint PMF table $p_{XY}(x, y)$
2. For each x_i, compute:

   |----------|

   $$p_X(x_i) = \sum_j p_{ij}$$

3. For each y_j, compute:

   |----------|

   $$p_Y(y_j) = \sum_i p_{ij}$$

4. Verify normalization conditions

2.2 Worked Example: Marginal PMF

Problem (Slide 5)

Let the joint PMF be:

	$y = 0$	$y = 1$
$x = 0$	$\frac{1}{8}$	$\frac{1}{4}$
$x = 1$	$\frac{3}{8}$	$\frac{1}{4}$

Concept Applied

Marginal PMF computation using summation over the other variable.

Solution

Step 1: Compute $p_X(x)$

|----------|

$$p_X(0) = \frac{1}{8} + \frac{1}{4}$$

|----------|

$$p_X(0) = \frac{1}{8} + \frac{2}{8} = \frac{3}{8}$$

|----------|

$$p_X(1) = \frac{3}{8} + \frac{1}{4}$$

|----------|

$$p_X(1) = \frac{3}{8} + \frac{2}{8} = \frac{5}{8}$$

Step 2: Compute $p_Y(y)$

|----------|

$$p_Y(0) = \frac{1}{8} + \frac{3}{8}$$

|----------|

$$p_Y(0) = \frac{4}{8} = \frac{1}{2}$$

|----------|

$$p_Y(1) = \frac{1}{4} + \frac{1}{4}$$

|----------|

$$p_Y(1) = \frac{2}{4} = \frac{1}{2}$$

Step 3: Final Result

|----------|

$$p_X(x) = \{3/8 \text{ at } x = 0,\ 5/8 \text{ at } x = 1\}$$

|----------|

$$p_Y(y) = \{1/2 \text{ at } y = 0,\ 1/2 \text{ at } y = 1\}$$

Key Logical Takeaways

- Marginals are obtained via summation across rows/columns
- Each marginal distribution satisfies normalization

Implied Mistakes

- Incorrect summation across wrong dimension (row vs column)
- Failure to convert fractions to common denominator

2.3 Correlation

(A) Definitions and Concepts

A basic way to measure joint behavior is through the product moment:

|----------|

$$R_{XY} = E[XY]$$

It tells the average value of the product XY. *(Slide 6)*

(B) Interpretation

- Depends on the scale of X and Y
- Computed about the origin

- Combines:
  - Mean values
  - Spread
  - Dependence

(C) Properties and Results

|----------|

$$R_{XY} = 0 \Rightarrow \textbf{orthogonal about the origin}$$

Not a clean measure of centered dependence *(Slide 7)*

(D) Transition

To remove the effect of means:

→ Center variables
→ Leads to covariance

2.4 Covariance

(A) Definitions and Concepts

|----------|

$$\mathrm{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

Where:

|----------|

$$\mu_X = E[X], \quad \mu_Y = E[Y]$$

Interpretation (Slide 8–9):

- Measures joint variation of centered variables
- Indicates direction of linear variation

(B) Derivation (Algebraic Form)

Start:

|----------|

$$\mathrm{Cov}(X,Y) = E[(X-\mu_X)(Y-\mu_Y)]$$

Expand:

|----------|

$$= E[XY - X\mu_Y - Y\mu_X + \mu_X\mu_Y]$$

Apply linearity:

|----------|

$$= E[XY] - \mu_Y E[X] - \mu_X E[Y] + \mu_X\mu_Y$$

Substitute:

|----------|

$$E[X] = \mu_X,\ E[Y] = \mu_Y$$

Thus:

|----------|

$$= E[XY] - \mu_X\mu_Y$$

Final:

|----------|

$$\mathrm{Cov}(X,Y) = E[XY] - E[X]E[Y]$$

(C) Properties and Results

1. **Symmetry**

|----------|

$$\mathrm{Cov}(X,Y) = \mathrm{Cov}(Y,X)$$

2. **Variance as Special Case**

|----------|

$$\mathrm{Cov}(X, X) = \mathrm{Var}(X)$$

3. **Linearity**

|----------|

$$\mathrm{Cov}(aX + b, cY + d) = ac\,\mathrm{Cov}(X, Y)$$

4. **Independence**

|----------|

$$X \perp Y \Rightarrow \mathrm{Cov}(X, Y) = 0$$

Statement:

Zero covariance means uncorrelated, but does not imply independence.

(D) Problem-Solving Methodology

1. Use:

   |----------|

   $$\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

2. Compute:
   - $E[XY]$
   - $E[X]$, $E[Y]$
3. Substitute values
4. Simplify

2.5 Worked Example 1 (Covariance)

Problem (Slide 12)

If:

|----------|

$$Y = -2X + 3$$

Find $\mathrm{Cov}(X, Y)$

Concept Applied

Covariance algebraic formula with linear transformation.

Solution

Start:

|----------|

$$\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

Since:

|----------|

$$Y = -2X + 3$$

Compute:

|----------|

$$XY = X(-2X + 3) = -2X^2 + 3X$$

|----------|

$$E[Y] = E[-2X + 3] = -2E[X] + 3$$

Substitute:

|----------|

$$\mathrm{Cov}(X, Y) = E[-2X^2 + 3X] - E[X](-2E[X] + 3)$$

Expand:

|----------|

$$= -2E[X^2] + 3E[X] + 2(E[X])^2 - 3E[X]$$

Simplify:

|----------|

$$= -2E[X^2] + 2(E[X])^2$$

Factor:

|----------|

$$= -2(E[X^2] - (E[X])^2)$$

Thus:

|----------|

$$\mathrm{Cov}(X, Y) = -2\mathrm{Var}(X)$$

Key Logical Takeaways

- Linear transformations scale covariance
- Variance identity used

Implied Mistakes

- Not expanding XY correctly
- Missing expectation linearity

2.6 Worked Example 2 (Continuous Case)

Problem (Slide 14)

|----------|

$$f_{XY}(x, y) = \frac{xy}{96}, \quad 0 < x < 4,\ 1 < y < 5$$

Compute:

1. $E[X]$

2. $E[Y]$
3. $E[XY]$
4. $E[2X + 3Y]$
5. $\mathrm{Var}(X)$
6. $\mathrm{Var}(Y)$
7. $\mathrm{Cov}(X, Y)$
8. ρ_{XY}

Solution

1. Compute $E[X]$

|----------|

$$E[X] = \int_0^4 x f_X(x) dx$$

|----------|

$$= \int_0^4 x \cdot \frac{x}{8} dx = \frac{1}{8} \int_0^4 x^2 dx$$

|----------|

$$= \frac{1}{8} \left[\frac{x^3}{3} \right]_0^4 = \frac{1}{8} \cdot \frac{64}{3}$$

|----------|

$$= \frac{8}{3}$$

---

2. Compute $E[Y]$

|----------|

$$E[Y] = \int_1^5 y \cdot \frac{y}{12} dy = \frac{1}{12} \int_1^5 y^2 dy$$

|----------|

$$= \frac{1}{12} \left[\frac{y^3}{3} \right]_1^5$$

|----------|

$$= \frac{1}{12} \cdot \frac{124}{3} = \frac{31}{9}$$

---

3. Compute $E[XY]$

Since X and Y are independent:

|----------|

$$E[XY] = E[X]E[Y]$$

|----------|

$$= \frac{8}{3} \cdot \frac{31}{9} = \frac{248}{27}$$

---

4. Compute $E[2X + 3Y]$

|----------|

$$E[2X + 3Y] = 2E[X] + 3E[Y]$$

|----------|

$$= 2\left(\frac{8}{3}\right) + 3\left(\frac{31}{9}\right)$$

|----------|

$$= \frac{16}{3} + \frac{31}{3} = \frac{47}{3}$$

---

5. Compute $\mathrm{Var}(X)$

|----------|

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2$$

|----------|

$$E[X^2] = \frac{1}{8}\int_0^4 x^3 dx = \frac{1}{8} \cdot 64 = 8$$

|----------|

$$\mathrm{Var}(X) = 8 - \left(\frac{8}{3}\right)^2 = \frac{8}{9}$$

---

6. Compute $\mathrm{Var}(Y)$

|----------|

$$E[Y^2] = \frac{1}{12}\int_1^5 y^3 dy = 13$$

|----------|

$$\mathrm{Var}(Y) = 13 - \left(\frac{31}{9}\right)^2 = \frac{92}{81}$$

---

7. Compute Covariance

|----------|

$$\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

|----------|

$$= \frac{248}{27} - \frac{248}{27} = 0$$

---

8. Compute Correlation

|----------|

$$\rho_{XY} = \frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}}$$

|----------|

$$\rho_{XY} = 0$$

Key Logical Takeaways

- Independence ⇒ $E[XY] = E[X]E[Y]$
- Covariance becomes zero
- Correlation becomes zero

Implied Mistakes

- Incorrect marginal extraction
- Incorrect limits of integration
- Missing independence condition

2.7 Pearson's Correlation Coefficient

(A) Definition

|----------|

$$\rho_{XY} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y}$$

|----------|

$$= \frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}}$$

(B) Properties

- Measures strength and direction
- Normalized covariance
- Unit-free

(C) Range and Interpretation

|----------|

$$-1 \leq \rho_{XY} \leq 1$$

|----------|

$$\rho_{XY} > 0 \Rightarrow \textbf{positive linear trend}$$

|----------|

$$\rho_{XY} < 0 \Rightarrow \textbf{negative linear trend}$$

|----------|

$$\rho_{XY} = 0 \Rightarrow \textbf{no linear correlation}$$

2.8 Joint Gaussian Random Variables

(A) Definition

A pair (X, Y) is jointly Gaussian if:

- Joint density has bivariate Gaussian form
- Marginals are Gaussian

(B) Algebraic Form

|----------|

$$f_{XY}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(\frac{-1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right)\right]\right)$$

(C) Parameters

- μ_X, μ_Y: means
- σ_X, σ_Y: standard deviations
- ρ: Pearson correlation coefficient

(D) Interpretation

- Used to model joint behavior of continuous variables
- Appears in:
  - Signal processing
  - Communication systems
  - Noisy measurements

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3. Final Revision Section

(A) Formula Compendium

|----------|

$$p_X(x) = \sum_y p_{XY}(x, y)$$

|----------|

$$p_Y(y) = \sum_x p_{XY}(x, y)$$

|----------|

$$R_{XY} = E[XY]$$

|----------|

$$\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

|----------|

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2$$

|----------|

$$\rho_{XY} = \frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}}$$

|----------|

Joint Gaussian PDF (as defined above)

(B) Core Concept Statements

- Marginal PMF is obtained by summing joint PMF
- Correlation $E[XY]$ is origin-based
- Covariance removes mean effects
- Zero covariance does not imply independence
- Pearson coefficient normalizes covariance
- Joint Gaussian defined by mean, variance, correlation

(C) Standard Solution Patterns

1. **Marginal PMF**
    - Sum rows and columns
2. **Covariance**
    - Compute $E[XY]$, $E[X]$, $E[Y]$
3. **Expectation (Continuous)**
    - Use definite integrals with limits
4. **Variance**
    - Use $E[X^2] - (E[X])^2$
5. **Correlation**
    - Normalize covariance

End of Lecture Scribe

Sources